Scatter-Limited Hybrid Monte Carlo, Deterministic Transport with Quasi–Monte Carlo Sampling

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Slides & paper: johanneskrotz.org/downloads

Transport model and why hybridize

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Linear Boltzmann / transport equation

$$\underbrace{\partial_t \Psi + \Omega \cdot \nabla_x \Psi}_{L\Psi} + \sigma_a \Psi = \sigma_s S[\Psi] + Q, \qquad S[\Psi] = \frac{1}{4\pi}\langle \Psi \rangle - \Psi$$

- $\Psi(x, \Omega, t)$ is the angular flux.
- Scalar flux: $\phi(x, t) = \langle \Psi \rangle = \int_{S^2} \Psi(x, \Omega, t)\, d\Omega$.

Numerical tension

- Monte Carlo resolves directional transport naturally, but has sampling noise.
- Deterministic S_N–DG is structured and efficient for smoother scattered fields, but angular coupling is expensive.
- Hybridization lets MC handle the early part and S_N–DG the smoother tail.

From 1st-collision to N_s-collision hybrid

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1st-collision hybrid

$$\Psi = \underbrace{\Psi_0}_{\text{MC}} + \underbrace{\Psi_{\geq 1}}_{S_N\text{–DG}}$$

$$L\Psi_0 + \sigma_t \Psi_0 = Q,$$
$$L\Psi_{\geq 1} + \sigma_t \Psi_{\geq 1} = \frac{\sigma_s}{4\pi}\langle \Psi_0 + \Psi_{\geq 1}\rangle.$$

one collision handled before hand-off

N_s-collision hybrid

$$\Psi = \underbrace{\sum_{n=0}^{N_s-1} \Psi_n}_{\text{MC}} + \underbrace{\Psi_{N_s}}_{\text{exactly } N_s} + \underbrace{\sum_{n\geq N_s+1} \Psi_n}_{S_N\text{–DG}}$$

$$L(\Psi_{\lceil N_s \rceil} + \Psi_{N_s}) + \sigma_a \Psi_{\lceil N_s \rceil} + \sigma_t \Psi_{N_s} = Q + \sigma_s S[\Psi_{\lceil N_s \rceil}],$$
$$L\Psi_{\lfloor N_s+1 \rfloor} + \sigma_t \Psi_{\lfloor N_s+1 \rfloor} = \sigma_s S[\Psi_{\lfloor N_s+1 \rfloor}] + \frac{\sigma_s}{4\pi}\langle \Psi_{N_s}\rangle.$$

MC up to N_s, $\quad N_s = 0$ recovers the 1st-collision split

Relabeling: $\quad L\Psi_R + \sigma_t \Psi_R = Q + \frac{\sigma_s}{4\pi}\langle \Psi_{\lceil N_s \rceil} + \Psi_{N_s} + \Psi_{\lfloor N_s+1 \rfloor}\rangle, \qquad \Psi(t_{n+1}) \leftarrow \Psi_R(t_{n+1})$

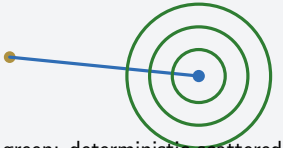
What changes in the split?

1st-collision hybrid

$$L\Psi_0 + \sigma_t \Psi_0 = Q$$

$$L\Psi_{\geq 1} + \sigma_t \Psi_{\geq 1} = \frac{\sigma_s}{4\pi} \langle \Psi_0 + \Psi_{\geq 1} \rangle$$

blue: MC to first collision



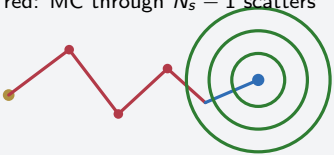
green: deterministic scattered tail

N_s-collision hybrid

$$L(\Psi_{\lceil N_s \rceil} + \Psi_{N_s}) + \sigma_a \Psi_{\lceil N_s \rceil} + \sigma_t \Psi_{N_s} = Q + \sigma_s S[\Psi_{\lceil N_s \rceil}]$$

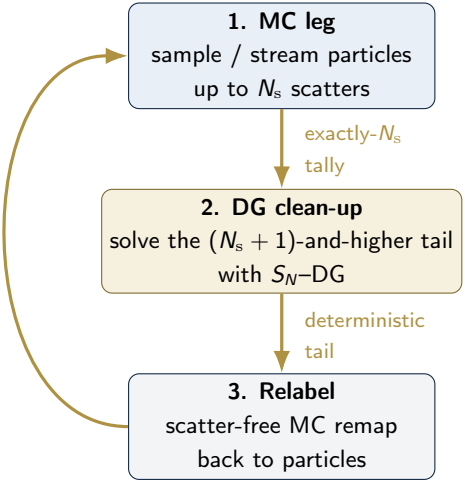
$$L\Psi_{\lfloor N_s+1 \rfloor} + \sigma_t \Psi_{\lfloor N_s+1 \rfloor} = \sigma_s S[\Psi_{\lfloor N_s+1 \rfloor}] + \frac{\sigma_s}{4\pi} \langle \Psi_{N_s} \rangle$$

red: MC through $N_s - 1$ scatters



blue: N_s hand-off, green: S_N tail

One step: MC, DG, and relabeling



Step 1: scatter-limited MC

$$L(\Psi_{\lceil N_s \rceil} + \Psi_{N_s}) + \sigma_a \Psi_{\lceil N_s \rceil} + \sigma_t \Psi_{N_s} = Q + \sigma_s S[\Psi_{\lceil N_s \rceil}]$$

MC advances particles only up to the scatter cap N_s.

Step 2: deterministic clean-up

$$L\Psi_{\lfloor N_s+1 \rfloor} + \sigma_t \Psi_{\lfloor N_s+1 \rfloor} = \sigma_s S[\Psi_{\lfloor N_s+1 \rfloor}] + \frac{\sigma_s}{4\pi}\langle \Psi_{N_s} \rangle$$

The exactly-N_s tally seeds the high-scatter tail solve.

Step 3: relabeling

$$L\Psi_R + \sigma_t \Psi_R = Q + \frac{\sigma_s}{4\pi}\langle \Psi_{\lceil N_s \rceil} + \Psi_{N_s} + \Psi_{\lfloor N_s+1 \rfloor} \rangle$$

Reset particles to birth data and carry the solution into the next step.

Relabeling: turn the full step back into an MC state

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Relabeling equation

$$L\Psi_R + \sigma_t \Psi_R = Q + \frac{\sigma_s}{4\pi}\langle \Psi_{\lceil N_s \rceil} + \Psi_{N_s} + \Psi_{\lfloor N_s+1 \rfloor}\rangle$$

$$\Psi(t_{n+1}) \leftarrow \Psi_R(t_{n+1})$$

Why relabel?

- Without relabeling, mass keeps drifting into the deterministic tail.
- Then long-time accuracy is increasingly set by the cleanup solve.
- Relabeling resets the next step to an MC representation.

How it is done in practice

1. Assume the full RHS is known from the MC tallies and the S_N–DG solve.
2. Run one *scatter-free* MC transport with attenuation σ_t.
3. Reuse the particles from step 1 by resetting them to their birth data.
4. Copy particles coming from initial/boundary data.

Practical point: no explicit resampling from the DG angular distribution is needed.

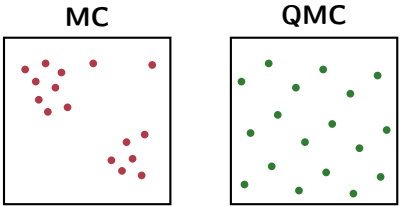
Why QMC can help here

General challenge

For fully scattering MC in heterogeneous media, the number of collision events is path dependent. That means the effective stochastic dimension is not fixed in advance, which makes classical QMC constructions awkward.

Why the hybrid helps

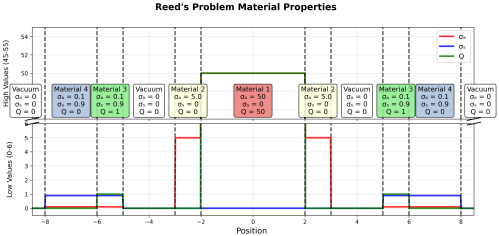
- The relabeling step is scatter-free.
- The implementation can keep the existing inverse-CDF maps and simply replace pseudorandom draws by Halton points.
- In the reported experiments, that change improved convergence in the hybrid setting with little extra cost.



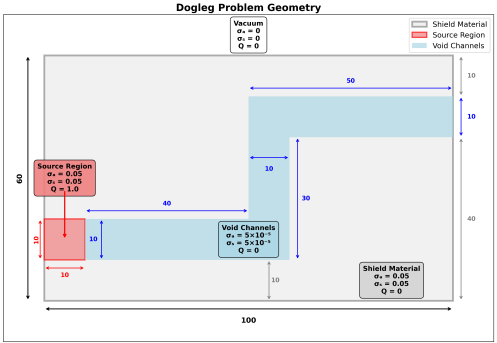
more even coverage
of the sampling space

Benchmark layouts

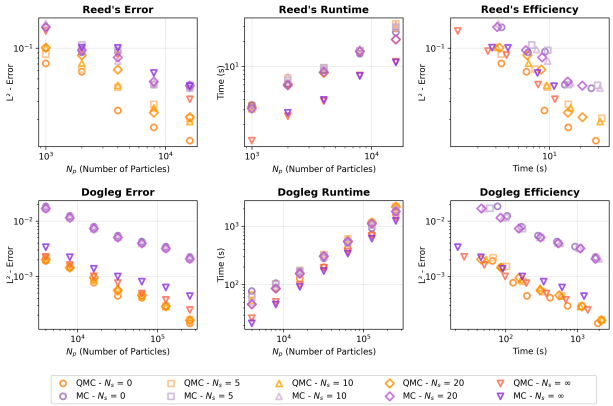
Reed's problem

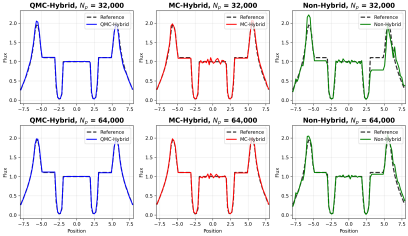


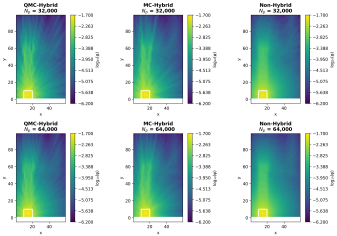
Dogleg problem



Benchmark results







QMC vs. MC: what is visible in Table I

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Convergence-rate and cleanup-iteration summary

	Reed			Dogleg		
N_s	SN iters	α_{QMC}	α_{MC}	SN iters	α_{QMC}	α_{MC}
0	31	-0.69	-0.50	7	-0.61	-0.49
5	26	-0.64	-0.51	4	-0.61	-0.49
10	21.4	-0.63	-0.53	1	-0.61	-0.49
20	12.2	-0.64	-0.49	1	-0.61	-0.49
∞	0	-0.56	-0.46	0	-0.53	-0.47

Takeaway

- Hybrid + QMC converges faster than hybrid + MC.
- The gain is strongest in the hybrid setting.

Role of N_s

- Larger N_s reduces the SN cleanup burden.
- Accuracy changes mildly; the work split changes.

Takeaways

1. The 1st-collision hybrid becomes an arbitrary scatter-limit hybrid.
2. MC handles the low-scatter, directional part; S_N–DG handles the high-scatter tail.
3. Relabeling is essential: it prevents the deterministic cleanup from taking over the long-time solution quality.
4. Replacing pseudorandom draws by Halton points gives a clear accuracy benefit in the hybrid setting at little extra cost.

One-sentence summary

The hybrid appears to create MC subproblems where QMC is much more effective in practice than in a comparable non-hybrid solve.

Outlook

- Can N_s be chosen adaptively in space, time, or by local optical thickness / variance indicators?
- How much larger is the payoff in a parallel implementation, where the MC leg is far easier to scale than the coupled cleanup solve?
- Could the cleanup stage be cheaper than S_N–DG in strongly scattering regions?
- The current results are still preliminary: two benchmarks, sequential runs, and one deterministic cleanup setting.

Questions?